

dimensions must correspond to the light's very short wavelength. We are able to fabricate antennae less than a micron in length."

To test the efficacy, researchers measured the antenna's ability to absorb and emit energy. By illuminating the antennae, the researchers were able to measure their ability to re-emit radiation efficiently, and determine the amount of power lost in the circuit—a simple matter of measuring the wattage going in and coming out.

By adapting classic metallic antennae to absorb light waves at optical frequencies, a much higher rate of conversion from light into usable energy could be achieved. Such efficiency, combined with a lower material cost, would mean a cost-effective way to harvest and utilise green energy.

Aerial robot system to save fire-fighters' lives

University of Cincinnati researchers have developed a system that integrates a small, unmanned aircraft with global positioning system (GPS), environmental data, video and fire-prediction software to give real-time information about where a fire is burning and where it is moving. The system is designed to assist fire-fighters overcome a major handicap during a wildfire—the inability to see the whole fire.

The system is built around a small, unmanned aerial vehicle from Marcus UAV Inc. The 1.9kg vehicle has a 1.4-metre wingspan and can fly faster than 56 km per hour to altitudes of 3048 metres on flights lasting approximately an hour. While in the air, it can transmit video as it navigates using GPS.

At the fire command centre, the inflight data is merged with Google Earth images, weather data and fire-prediction software to help the incident commander make informed and effective decisions.

Face-reading lie detector

Researchers at Bradford and Aberystwyth universities are developing a non-invasive lie-detection test based on facial images. The computerised

system uses a simple video camera, a high-resolution thermal imaging sensor and a suite of algorithms.

Traditional polygraph uses measures such as skin conductivity and blood pressure of the subject with a set of wires attached to his skin. In the new system, the video camera and thermal imager record changes in the facial symmetry of the subject. The combination allows lying to be detected with 90 per cent accuracy even in unwilling subjects.

The infrared camera detects temperature changes of as little as 0.1 degree, especially in the skin around the eyes. Because lying involves increased brain activity, blood flow in the face is a key indicator of dishonesty.

The video camera is used to compare the subject's facial expressions under questioning with their appearance during a series of harmless questions, which the subject will answer truthfully.

"We use these questions to establish a baseline condition for the face. Different people have different levels of nervousness. Establishing a baseline allows us to see whether the person is emotional, aroused or nervous. Some of the key indicators are swallowing, lip biting and an increase in facial asymmetry," said Professor Hassan Ugail of the University of Bradford.

Robot with a human face

A team at the Institute for Cognitive Systems (ICS) at TU München, Germany, in collaboration with a group in Japan has developed the prototype of a talking robot head.

Although 3D heads have been developed earlier, these are more cartoon-like. Mask-bot, however, can display realistic 3D heads on a transparent plastic mask, and change the face on-demand. A projector behind the mask accurately beams a human face onto the back of the mask, creating very realistic features that can be seen from various angles, including the side.

Many comparable systems project faces onto the front of a mask—following the same concept as cinema projection. However, Mask-bot uses on-board rear

projection to ensure a seamless face-to-face interaction.

To replicate facial expressions, Takaaki Kuratate developed a talking-head animation engine—a system in which a computer filters an extensive



Talking-head robot

series of face motion data collected by a motion-capture system and selects the facial expressions that best match a specific sound, called a phoneme, when it is being spoken. The computer extracts a set of facial coordinates from each of these expressions, which it can then assign to any new face, thus bringing it to life. Emotion synthesis software delivers the visible emotional nuances that indicate, for example, when someone is happy, sad or angry.

Anti-humidity PCBs

Gigabyte has introduced motherboards using its proprietary glass-fabric PCB technology. These motherboards are capable of fighting moisture caused by humidity.

Humidity is a major cause for short circuits and system malfunction. The new PCB material reduces the space between fibre weaves. This reduction of space acts as an anti-humidity shield for the fibre weave and makes it difficult for moisture to penetrate, compared to conventional motherboard PCBs.

In addition to protection of the motherboard at PCB level, certain improvements have been made in the actual PCB structure. The PCB weave of these motherboards is thinner and flatter, which further prevents moisture penetration and eventual shorting. The technology has been implemented in all the recently launched Ultra Durable 4 Classic motherboards.

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